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Jointly Modeling Means and Variances for Nonlinear Mixed Effects Models with Measurement Errors and Outliers

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Summary: In the longitudinal data analysis, the within-individual repeated measurements often exhibit large variations and these variations appear to change over time. A good understanding the nature of the within-individual systematic and random variations allows us to conduct more efficient statistical inferences and make better predictions. Motivated by HIV viral dynamic studies, we considered a nonlinear mixed effects (NLME) model for modeling the longitudinal means, together with a model for the within-individual variances which also allows us to address outliers in the repeated measurements. Statistical inference was then based on a joint model for the mean and variance, implemented by a computationally efficient approximate method. Extensive simulations evaluated the proposed method. We found that the proposed method produces more efficient estimates than the corresponding method without modeling the variances. Moreover, the proposed method provides robust inference against outliers. The proposed method was applied to a recent HIV-related dataset, with interesting new findings.

Key words: h-likelihood, joint model, measurement error, robust.

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1. Simulation Studies

In this section, we evaluated the proposed methods, i.e., the JM and TS methods, and compared them with the commonly used LB method via extensive simulations. Note that the JM and TS methods model the within-individual variance while the LB method does not.

Data were simulated for $n = 100$ subjects, with $n_i = 15$ repeated measurements for each subject. The entire study period was scaled to be $[0, 1]$, with n_i repeated measuring times, $t_{ij}, j = 1, \cdots, n_i$, equally spaced. The longitudinal responses y_{ij} were generated based on the following (mean) model

$$y_{ij} = (\beta_1 + u_{1i}) + \beta_2 e^{-(\beta_3 + u_{3i})t_{ij}} + e_{ij}, \qquad i = 1, \cdots, n, \quad j = 1, \cdots, n_i, \tag{1}$$

where the random effects $\boldsymbol{u}_i = (u_{1i}, u_{3i}) \sim N(0, D)$ with $D = \begin{pmatrix} d_1^2 & \rho d_1 d_3 \\ \rho d_1 d_3 & d_3^2 \end{pmatrix}$, the within-individual errors $e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$. The variance σ_{ij}^2 were generated based on the variance model

$$\log(\sigma_{ij}^2) = \alpha_0 + \alpha_1 z_{ij}^* + a_i. \tag{2}$$

We considered two cases in simulating data: data without outliers and data with outliers. These two cases correspond to two assumed distributions for the random effect a_i in the variance model:

- Setting I: base setting – data without outliers, which corresponds to simulating data with $a_i \sim \mathcal{N}(0, \sigma_a^2)$,
- Setting II: robust setting – data with outliers, which corresponds to simulating data with $\exp(a_i) \sim k/\chi_k^2$, i.e., the response data were simulated based on the $t(k)$ distribution (but in model fitting the data were still assumed to be normally distributed).

In Setting II, We considered two values for the degrees of freedom k in the inverse χ^2 distribution: $k = 3$ and $k = 5$. In this setting, note that outliers in the data were naturally

simulated by the corresponding heavy-tail t_3 and t_5 distributions. The error-prone and time-varying covariate z_{ij} and its true value z^*_{ij} were simulated based on the following measurement error model

$$z_{ij} = z^*_{ij} + \varepsilon_{ij} \equiv \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t^2_{ij} + b_i + \varepsilon_{ij}, \tag{3}$$

for $i = 1, \cdots, n$, $j = 1, \cdots, n_i$, where $b_i \sim \mathcal{N}(0, \sigma_b^2)$ and $\varepsilon_{ij} \sim \mathcal{N}(0, \omega^2)$.

The true parameter values were chosen based on modelling fitting for a real data set. Specifically, we set true parameter values as follows

$$\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3)^T = (2.5, 3.0, 7.5)^T,$$

$$\boldsymbol{\alpha} = (\alpha_0, \alpha_1)^T = (-8, 1.5)^T,$$

$$\boldsymbol{\gamma} = (\gamma_0, \gamma_1, \gamma_2)^T = (5.2, 1.6, -1.2)^T,$$

$$(d_1, d_3) = (0.4, 1.0), \quad \rho = 0.5, \quad \sigma_a = 0.7, \quad \sigma_b = 0.4, \quad \omega = 0.2.$$

The random effects were assumed to be independent across the three different models (mean model, variance model, and measurement error model). Each model parameter was estimated using three methods JM, TS, and LB. The starting values for the parameters were set to be their estimates from the LB method (as there are no LB estimates for $\boldsymbol{\alpha}$, their true values were used), and the number of simulation repetition was set to be $M = 200$.

For each parameter and each method, we reported "estimate" (average of the M estimates from the M simulations), "SE.mo" (average of the M model-based SE's), "SE.em" (sample or empirical SE of the M estimates), and percent "rBias" and percent "rMSE" which were the relative bias and relative mean square error (MSE) of the M parameter estimates defined as follows (say, for parameter β):

$$\text{rBias (\%) of } \hat{\beta} = \frac{1}{M} \sum_{m=1}^{M} \left(\hat{\beta}^{(m)} - \beta \right) / |\beta| \times 100\%,$$

$$\text{rMSE (\%) of } \hat{\beta} = \sqrt{\frac{1}{M} \sum_{m=1}^{M} \left(\hat{\beta}^{(m)} - \beta \right)^2} / |\beta| \times 100\%,$$

where $\hat{\beta}^{(m)}$ was the estimate of β in the m-th simulation run. We also reported "Coverage", which was the empirical coverage of the true parameter value based on the M 95% confidence intervals from simulations. In addition, the *parametric* bootstrapping method was also used to compute the SE for the parameter estimates in the JM method, denoted by "SE.bt", with corresponding coverage denoted by "Coverage.bt". We conducted 100 bootstrapping runs for each of the M simulation runs.

To evaluate the prediction performance, we also examine the coverage of the simulated future data for each of the three methods. More specifically, besides the $n_i = 15$ equally-spaced repeated measurements, we simulated three more data points for each subject, that is the 16th, 18th and 20th measurements, to represent the future data. For each method, we calculated the predicted values and standard errors for those future data based on the estimated parameters, and then obtained the empirical coverage of the true data point values based on the 95% confidence intervals. The average coverage rate for each future data point from the M simulation runs was reported. Additionally, for the parametric bootstrapping in the JM method, the 95% confidence interval was constructed by finding the empirical 2.5% and 97.5% percentiles among the 100 predicted values from the 100 bootstrapping runs.

Table 1 presents the simulation results for the base setting where the simulated data did not have outliers for 20 simulation runs, as an preliminary illustration. Table 2 shows the predictive power of the three methods.

[Table 1 about here.]

[Table 2 about here.]

Table 1
Simulation results for the base setting without outliers

Method	Parameter	β_1	β_2	β_3	α_0	α_1	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
LB	Estimate	2.51	3.20	8.38	-	-	-	-	-
	SE.mo	0.07	0.48	1.54	-	-	-	-	-
	SE.em	0.06	0.43	1.63	-	-	-	-	-
	rBias(%)	0.48	6.56	11.74	-	-	-	-	-
	rMSE(%)	2.50	15.29	24.24	-	-	-	-	-
	Coverage	1.00	0.95	1.00	-	-	-	-	-
TS	Estimate	2.50	3.15	8.01	-8.40	1.55	5.21	1.61	-1.21
	SE.mo	0.05	0.21	0.71	0.51	0.09	0.04	0.08	0.07
	SE.em	0.07	0.44	2.16	0.71	0.12	0.06	0.10	0.09
	rBias(%)	-0.13	4.87	6.82	-4.99	3.47	0.11	0.72	-1.15
	rMSE(%)	2.77	15.19	28.83	9.96	8.77	1.09	6.00	7.00
	Coverage	0.80	0.95	0.80	0.80	0.85	0.90	0.85	0.90
JM	Estimate	2.48	3.05	7.48	-7.97	1.48	5.20	1.62	-1.22
	SE.mo	0.05	0.20	0.70	0.50	0.09	0.04	0.07	0.07
	SE.em	0.08	0.26	1.23	0.69	0.12	0.06	0.10	0.08
	SE.bt	0.06	0.24	0.86	0.72	0.13	0.04	0.08	0.07
	rBias(%)	-0.75	1.51	-0.23	0.33	-1.59	0.08	0.99	-1.34
	rMSE(%)	3.20	8.59	15.99	8.45	7.97	1.09	5.99	6.98
	Coverage	0.80	0.95	0.80	0.80	0.80	0.90	0.85	0.90
	Coverage.bt	0.85	0.95	0.85	0.95	1.00	0.90	0.85	0.90

LB: the linearization method based on **?**. *TS: two-step method. JM: Joint model.*

Table 2
Coverage rates of future data points for the base setting without outliers

Method	Future data points	16th	18th	20th
LB	Coverage	0.95	0.97	0.98
TS	Coverage	0.82	0.82	0.81
JM	Coverage	0.78	0.77	0.77
	Coverage.bt	0.83	0.86	0.89

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